Statistical Thinking for Data Science

Week 9: From a sample to a claim

How far the truth can be from what you measured

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First Semester, 2026/2027

Week 9: From a sample to a claim

Learning Objectives

By the end of this week you should be able to:

- explain what a sampling distribution is, and distinguish it from the distribution of the data;
- compute a standard error for a mean and for a proportion, and say what it measures;
- state the central limit theorem and the conditions under which it helps;
- construct and interpret a confidence interval, and avoid the standard misinterpretation;
- choose a sample size for a target margin of error, including the finite population correction;
- produce a bootstrap interval for a statistic with no formula, such as a median.

9.1 The question of the week

You measured 25 core samples and the mean grade was 2.41 g/t. The deposit does not care what your 25 samples said: it has a true mean grade you will never see. How far from it could your 2.41 plausibly be?

Everything in this chapter exists to answer that. Note what the question is not: it is not about how spread out the grades are. It is about how spread out *the mean of 25 grades* would be if you went out and did the whole exercise again.

Three distributions, kept separate

Population: all possible core samples, with mean μ and sd σ. **Sample**: the 25 you measured, with mean $\bar{x}$ and sd s, which you can plot as a histogram. **Sampling distribution**: the distribution of $\bar{x}$ over all the samples of 25 you might have taken. You never see it, you reason about it. Confusing the first and third is the commonest error in applied statistics.

9.2 The standard error

If observations are independent with standard deviation σ, the sample mean has

$$\mathrm{E}(\bar{X}) = \mu, \qquad \mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

The mean is unbiased: it is centred on the truth. Its spread shrinks with $\sqrt{n}$, not with n, which is the single most consequential fact about sampling.

n	9	36	144
$\mathrm{SE} = 0.6/\sqrt{n}$ (g/t)	0.200	0.100	0.050

Four times the samples for half the error. Sixteen times for a quarter. Every budget conversation about sample size is a conversation about this square root, and it is why a huge sample is not always worth its cost.

For a proportion, the same idea with $\sigma = \sqrt{p(1-p)}$:

$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}},$$

estimated by putting $\hat{p}$ in place of p.

Standard deviation or standard error?

The standard deviation describes the *data*: how much one borehole differs from another. The standard error describes an *estimate*: how much the mean of 25 boreholes would move if you repeated the study. Reporting one when you mean the other misstates your precision by a factor of $\sqrt{n}$, and both appear in reports written as "±" with no label.

9.3 The central limit theorem

For almost any population with a finite variance, the sampling distribution of $\bar{X}$ approaches a normal distribution as n grows, whatever the shape of the population.

Three practical qualifications:

- **How large is large?** For mildly skewed data, $n = 30$ is usually plenty. For strongly skewed data such as incomes or repair times, you may need hundreds, and the usual check is to bootstrap and look.
- **It is about the mean, not the data.** The theorem never says your observations become normal. Grades stay skewed no matter how many you take.
- **Independence is required.** Cluster samples, repeated measurements on the same truck, or time series with trends all break it, and the real standard error is larger than the formula says.

9.4 Confidence intervals

A 95% confidence interval for a mean, when σ is unknown and estimated by s, is

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}},$$

with t^* from the t distribution on $n - 1$ degrees of freedom. For a proportion,

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}},$$

with $z^* = 1.96$ for 95%, 1.645 for 90% and 2.576 for 99%.

Example 1: grade from 25 cores

$n = 25$, $\bar{x} = 2.41$ g/t, $s = 0.62$ g/t.

Solution

SE $= 0.62/\sqrt{25} = 0.124$. With 24 degrees of freedom, $t^* = 2.064$, so the margin is $2.064 \times 0.124 = 0.256$ and the interval is

$$2.41 \pm 0.26 \;=\; (2.15,\; 2.67) \text{ g/t}.$$

Saying it properly

"The 25 cores give an estimated mean grade of 2.41 g/t, with a 95% confidence interval from 2.15 to 2.67 g/t." Not: "there is a 95% probability the true grade is between 2.15 and 2.67". The truth is fixed; it is the interval that is random. The 95% refers to the procedure: intervals built this way capture the truth 95% of the time in the long run.

Example 2: a survey proportion

In a survey of 200 students, 84 say they have used the course tutor, so $\hat{p} = 0.42$.

Solution

SE $= \sqrt{0.42 \times 0.58/200} = 0.0349$, margin $1.96 \times 0.0349 = 0.068$, interval $(0.352,\; 0.488)$.

So "42% of students, give or take seven percentage points". If a colleague claims usage has reached half the class, this sample cannot refute it: 0.50 is just outside the interval, which is worth saying out loud rather than hiding.

9.4.1 What the interval does and does not include

The interval covers uncertainty from random sampling only. It says nothing about
- people who did not respond, who are usually different from those who did;
- a sampling frame that missed part of the population;
- measurement error, leading questions or miscoded data;
- the fact that you asked this question only after seeing the data.

In most real studies these dwarf the sampling error. A narrow interval around a biased estimate is precision about the wrong number, and that is the dangerous case because it looks rigorous.

9.5 Choosing a sample size

Fix the margin of error m you can live with and solve. For a proportion, using $p = 0.5$ because that is the worst case,

$$n = \frac{(z^*)^2\, p(1-p)}{m^2} = \frac{1.96^2 \times 0.25}{0.03^2} = 1067.1 \to 1068.$$

If the population is small, the finite population correction applies:

$$n_{\text{adj}} = \frac{n_0}{1 + (n_0 - 1)/N}.$$

For a faculty of $N = 1200$ students, $n_{\text{adj}} = 1067.1/(1 + 1066.1/1200) = 565.1$, so 566 students. Sampling nearly half a small population buys what a thousand would buy from a large one.

Before you ask for a bigger sample

Halving the margin needs four times the data. If the design is biased, a bigger sample makes the estimate more precisely wrong. Spend the money on non-response follow-up or better measurement first, and on sample size second.

9.6 The bootstrap: when no formula exists

There is a formula for the standard error of a mean. There is none you want to remember for a median, a trimmed mean, a ratio of two medians, or the output of a model. The bootstrap gets the answer by resampling.

1. Draw n observations from your sample *with replacement*.
2. Compute the statistic on that resample.
3. Repeat a few thousand times.
4. The spread of those values estimates the sampling distribution; the 2.5th and 97.5th percentiles give a 95% interval.

Example 3: a bootstrap interval for the turbidity data

The fifteen turbidity readings from the tutorial sheet: 12, 14, 15, 15, 16, 17, 18, 19, 21, 22, 24, 27, 31, 38, 46.

Solution

The sample mean is 22.33 and $s = 9.60$, so the formula gives SE $= 9.60/\sqrt{15} = 2.48$. Ten thousand bootstrap resamples give a bootstrap standard error of 2.39 and a 95% percentile interval of about (18.0, 27.3), which is close to the formula and slightly asymmetric because

the data are right skewed.
Now the median, for which no simple formula exists. The sample median is 19, and the bootstrap interval is (15, 24). It is wider in relative terms and lumpy, because a median from fifteen numbers can only take a few values.

Why bother

The bootstrap is not magic: it assumes your sample is representative, and it cannot repair a biased design. What it does is free you from formula hunting, so you can report an interval for the statistic that actually answers the question instead of the one with the convenient algebra.

9.7 Reporting an estimate

Four things, every time: the estimate, the interval, the sample size, and the source of the data. "42% (95% CI 35 to 49), $n = 200$ students who answered the class survey in week 4" is a complete sentence. "42%" is not. Round the interval to the precision the data support: an interval of $(0.3516, 0.4884)$ claims four decimal places from 200 people.

9.8 Doing it on a computer

R: `t.test(x)` prints the interval; `prop.test` or `binom.test` for proportions; `boot` or a four line `replicate(10000, mean(sample(x, replace = TRUE)))` for the bootstrap. Python: `scipy.stats.t.interval`, `statsmodels.stats.proportion.proportion_confint`, and `numpy.random.default_rng().choice` for resampling. Set a seed and report the number of resamples.

9.9 Glossary

Sampling distribution	the distribution of a statistic over repeated samples
Standard error	the standard deviation of a statistic's sampling distribution
Central limit theorem	the sample mean tends to normality as n grows
Confidence interval	an interval built by a procedure that captures the truth a stated proportion of the time
Margin of error	half the width of the interval
Finite population correction	the factor that shrinks the needed sample when the population is small
Bootstrap	resampling your own data with replacement to approximate a sampling distribution

9.10 Summary

Week 9 in six lines

1. The sampling distribution is about the estimate, not the data.
2. $\text{SE} = \sigma/\sqrt{n}$: four times the data halves the error.
3. The central limit theorem makes the mean normal, not the observations.
4. A confidence interval describes the procedure, not the probability of the truth.
5. Intervals cover random sampling error only, and bias is usually bigger.
6. When there is no formula, bootstrap, and say how many resamples you used.

Exercises

1. Ore grades have $\sigma = 0.6$ g/t. Find the standard error of the mean for $n = 9$, 36 and 144. How many samples are needed for a standard error of 0.02, and comment on whether that is worth paying for.

2. A sample of 25 cores gives $\bar{x} = 2.41$ and $s = 0.62$ g/t. Construct a 95% confidence interval, then state in one sentence what the 95% refers to, and in one sentence what it does not.

3. In 200 surveys, 84 students report using the tutor. Give a 95% interval for the proportion. Your head of department claims half the class uses it. What do you say?

4. You need a margin of error of 3 percentage points at 95% confidence. Find the sample size for a large population, then for a faculty of 1 200, then for a class of 150.

5. Explain to a colleague why their survey of 2 000 people recruited from one WhatsApp group is not more trustworthy than a random sample of 300, even though the interval is narrower.

6. Take the fifteen turbidity readings from the Week 1 tutorial. Compute the formula interval for the mean, then a bootstrap interval with 10 000 resamples, and explain the difference between them.

7. Bootstrap a 95% interval for the median of the same data, and say why no simple formula was used.

8. A report states "mean downtime 4.2 hours ± 1.1". List the three things the reader cannot tell from that sentence, and rewrite it properly.

Answers to the odd-numbered exercises

1. $0.6/\sqrt{9} = 0.200$, $0.6/\sqrt{36} = 0.100$, $0.6/\sqrt{144} = 0.050$ g/t. For SE $= 0.02$, $n = (0.6/0.02)^2 = 900$ samples. Whether that is worth paying for depends on whether a decision changes at a precision of 0.02 g/t; usually it does not, and the money goes to drilling elsewhere.

3. $\hat{p} = 0.42$, SE $= 0.0349$, interval $(0.352, 0.488)$. The claim of 50% lies just outside the interval, so the data are mildly inconsistent with it, but with 200 students you should say "our estimate is 42%, 35 to 49, so a true figure of 50% is unlikely though not excluded", not "the claim is false".

5. Both the narrow interval and the wide one are computed assuming random sampling. One WhatsApp group is a self-selected cluster, so the estimate is biased by whoever joins that group, and the interval measures only the random part of the error. A random sample of 300 has a wider interval around a number that is at least aimed at the right target.

7. The median is 19 and a bootstrap percentile interval is about $(15, 24)$. No formula was used because the sampling distribution of a median depends on the density of the population near the middle, which you do not know; resampling estimates it from the data you have.

Before next week

Bootstrap something of your own: any statistic, from any dataset you have, with 10 000 resamples, and bring the histogram of the bootstrap distribution. Next week we turn estimates into decisions, and meet the p-value, the most misused number in science.